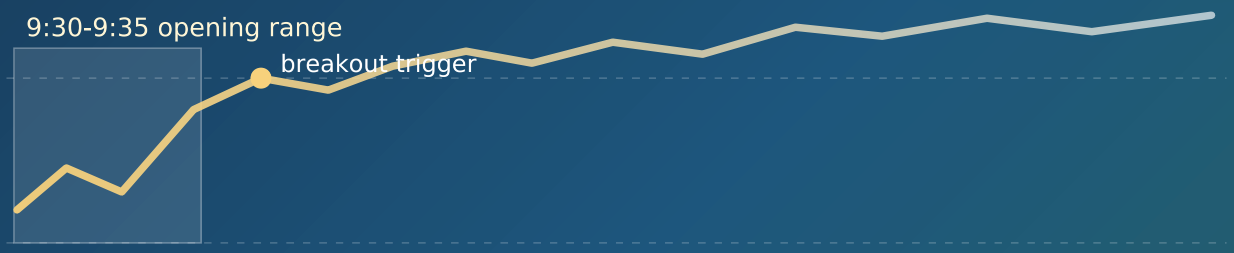
ADVANCED TRADING STRATEGY PLAYBOOK

Opening Range Breakout for Stocks in Play

A professional implementation manual for Strategy #2: catalyst-driven, relative-volume-filtered, risk-controlled ORB execution during the U.S. equity market open.

U.S. Equities | 9:30-10:15 a.m. ET | 5-Minute Opening Range Default | Advanced / Expert Traders



Core idea: trade only when opening price discovery, catalyst quality, and abnormal volume align.

Prepared: May 2026

Scope: Educational research and process playbook, not personal financial advice

Operational Scope and Risk Notice

This playbook converts the Opening Range Breakout, or ORB, into a repeatable professional workflow. It is designed for advanced and expert traders who already understand intraday execution, short selling, bracket orders, VWAP, ATR, liquidity, slippage, and risk-unit accounting.

SCOPE

- ☐ Primary market: U.S. listed common stocks during regular trading hours.
- ☐ Default range: first 5 minutes after the 9:30 a.m. ET open.
- ☐ Default window: scan at 9:35 a.m.; execute mostly 9:35-10:15 a.m.
- ☐ Default universe: liquid stocks above minimum price, ATR, and dollar-volume thresholds.
- ☐ Holding period: intraday only unless separately tested.

RISK NOTICE

Day trading has substantial risk. ORB trades concentrate risk in the most volatile part of the day. The evidence base is historical and cannot guarantee future performance. Realized performance depends on data quality, order handling, short-borrow access, slippage, fees, liquidity, and discipline.

Non-negotiable: no strategy in this playbook should be traded live without instrument-specific backtesting, simulated execution review, and a written kill-switch plan.

> **Core thesis:** ORB is not a generic chart pattern. The edge appears strongest when the opening range acts as a measurable expression of catalyst-driven order flow in a stock that is genuinely "in play." The trade is a momentum trade only when opening price discovery, relative volume, and risk/reward line up.

What this playbook intentionally excludes

Excluded area	Reason
Holding overnight by default	The playbook is designed as an intraday execution manual. Overnight exposure changes the risk model.
Low-float microcap momentum	The halt, dilution, spread, and manipulation risks require a separate microcap-specific framework.
Zero-DTE options ORB	Options introduce IV, Greeks, spread, assignment, and broker-specific execution issues.
Blind ORB on every ticker	The research edge is strongest when constrained to stocks with abnormal opening volume and catalyst relevance.

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1. Research Base and Edge Hypothesis

The most useful evidence does not say "all breakouts work." It says that a narrow, disciplined ORB implementation improves meaningfully when it focuses on liquid stocks with abnormal opening volume and a real reason to move.

ACADEMIC FINDING

Stocks in Play matter

Zarattini, Barbon, and Aziz analyzed more than 7,000 U.S. stocks from 2016 to 2023 and focused on 5-minute ORB strategies. Their abstract reports that limiting trades to the top 20 Stocks in Play produced more than 1,600% total net performance, a Sharpe ratio of 2.81, and annualized alpha of 36%, versus 198% for passive S&P 500 exposure over the same period.[1]

IMPLEMENTATION EVIDENCE

Relative-volume filter

A QuantConnect recreation defines the universe as liquid U.S. equities above $5 and above an ATR threshold, ranks Stocks in Play by first-5-minute volume divided by average first-5-minute volume over the previous 14 sessions, then trades the top candidates after the 9:35 a.m. scan.[2]

MARKET STRUCTURE

Why the open is special

Hong and Wang's market-closure model explains why information accumulation during a closure can create high opening volume, and why intraday volume and volatility can be higher around open and close than midday.[3]

BEHAVIORAL MECHANISM

Attention creates flow

Barber and Odean find that individual investors are net buyers of attention-grabbing stocks, including stocks in the news, stocks with abnormal trading volume, and stocks with extreme one-day returns.[4] That does not make every attention stock a buy; it explains why attention can concentrate flow.

> **Interpretation discipline:** The ORB literature is promising but not a license to overtrade. The practical edge is a conditional edge: catalyst + abnormal volume + executable liquidity + controlled risk + no disqualifying context.

Research-to-trading translation

Research observation	Trading implication	Playbook rule
Performance improves when restricted to Stocks in Play.	Selection matters more than the candle pattern.	Never trade ORB without a catalyst or abnormal first-5-minute relative volume.
Opening range is the price-discovery zone.	Do not predict before the range forms.	No entry before the first range completes unless you have a separate tested strategy.
Opening volume and volatility are elevated.	Stops must account for noise and slippage.	Use ATR-aware sizing, stop-limit controls, and max daily R loss.
Attention can concentrate trading in certain names.	Attention is fuel, not proof.	Rank attention names, then require price confirmation and liquidity.

2. ORB Edge Anatomy

The setup works only when the open reveals a directional imbalance that can persist. Professional traders should think in terms of market participants, not in terms of lines on a chart.

1

Catalyst

Earnings, guidance, macro shock, FDA decision, analyst action, index/sector event, or credible company-specific news changes the fair-value debate.

2

Abnormal volume

First-5-minute volume relative to its own historical first-5-minute baseline confirms participation. Raw volume alone is not enough.

3

Opening range

The first range defines who is trapped, where liquidity sits, and what price must break before momentum traders and stop orders engage.

4

Continuation

Continuation requires price holding outside the range, VWAP alignment, and market or sector support. A wick through the level is not enough.

5

Defined risk

The opening range provides a natural invalidation zone. Trade size is derived from distance to invalidation, not conviction.

6

Failure recognition

If price re-enters the range and accepts inside it, the breakout thesis is wrong. Expert execution cuts quickly.

ORB is a three-part bet

```
Expected Edge = Selection Edge x Confirmation Edge x Execution Edge

Selection Edge     = catalyst quality + abnormal relative volume + liquidity
Confirmation Edge  = breakout acceptance + VWAP/market/sector alignment
Execution Edge     = spread discipline + risk-unit sizing + slippage control
```

> **Professional mindset:** You are not buying the high of a candle. You are buying evidence that the market is still repricing the stock and that trapped or under-positioned participants may need to chase.

False-break versus true-break characteristics

Attribute	True-break candidate	False-break candidate
Catalyst	Material, current, easy to understand	Stale, vague, social-media driven, or already fully priced
First 5-min volume	Top-ranked relative to its own normal first 5-min volume	Only raw volume because float is tiny or spread is huge
Opening candle	Strong body, closes near extreme, range not excessive	Large wicks, indecision, or range wider than reasonable R/R
VWAP	Price breaks and holds on the correct side	Breaks level but immediately loses VWAP
Market/sector	Sector confirms or stock has exceptional relative strength	Broad tape is reversing against the trade

3. Daily Trading Timeline

The ORB edge is time-of-day specific. Most mistakes come from entering too early, chasing too late, or trading a ticker that was never truly in play.

7:00-8:30	**Market map.** Note overnight futures, macro calendar, mega-cap earnings, sector themes, and premarket liquidity. Build a preliminary catalyst list.
8:30-9:15	**Watchlist scoring.** Rank catalyst quality, gap size, premarket volume, range, borrow availability, spread, and prior-day levels. Remove untradeable names.
9:15-9:29	**Plan orders, do not predict.** Mark premarket high/low, prior close, prior high/low, VWAP expectation, ATR, and maximum risk. No FOMO entries.
9:30-9:35	**Observe the opening range.** Let the first 5-minute range form. Record ORH, ORL, ORW, candle body, wick ratio, first-5-minute volume, and RVOL5.
9:35-9:45	**Primary execution window.** Trade A+ breaks only. If no clean trigger occurs, wait. The first valid break is the cleanest expression of the strategy.
9:45-10:15	**Secondary window.** Prefer retests, pullback continuations, or failed-break traps. Do not chase extended names without fresh consolidation.
10:15-11:00	**Degrade signal quality.** New ORB entries require stricter confirmation. Manage existing winners; cut stalled trades.
After 11:00	**No default new ORB.** If trading later, it is now a different strategy: VWAP continuation, intraday trend, or mean reversion.

Best-case timing

The ideal long trade triggers shortly after 9:35, holds above ORH and VWAP, does not violate the breakout level on retest, and reaches at least +1R before 10:00.

Worst-case timing

The low-quality trade triggers late, far above VWAP, after a wide opening candle, into a resistance level, while the sector and market are rolling over.

4. Universe Construction and Scanner

Professional ORB begins before the open. The scanner should identify stocks where the first 5 minutes are likely to matter and where execution quality is acceptable.

Baseline universe

Filter	Default playbook threshold	Reason
Price	Greater than $5, preferably greater than $10 for discretionary trading	Reduces tick-size distortions, spread risk, and penny-stock behavior.
Average dollar volume	Top liquid universe; practical minimum $20M-$50M ADV for most traders	Allows entry and exit without becoming the market.
ATR14	Greater than $0.50, or sufficient for your fee/slippage structure	ORB needs range; extremely low ATR names do not pay for execution risk.
Borrow	Confirmed easy-to-borrow or located before short trigger	Short ORB signals are irrelevant if the trader cannot borrow shares.
Spread	Prefer spread below 5-10 bps in large caps; use stricter limits for size	Wide spreads destroy positive expectancy and distort stops.
Halts/news	No unresolved halt, pending binary event, or unreliable news source	Gap-through-stop and reopening risk can invalidate the model.

Stocks-in-play ranking formula

```
RVOL5 = Volume from 9:30:00 to 9:34:59 today
        / Average volume from 9:30:00 to 9:34:59 over prior 14 sessions

Stock-in-Play Rank = descending RVOL5 among eligible liquid stocks
```

The QuantConnect recreation of the 2024 study uses a similar first-5-minute relative-volume concept and selects the highest relative-volume stocks from a liquid universe.[2]

Institutional-quality setup

Real catalyst, high dollar volume, tight spread, strong RVOL5, clean technical level.

Tradable but reduced size

Good catalyst and range, but spread, borrow, or sector context is less clean.

Exclude

Microcap promotion, no borrow, massive spread, pending halt, unreliable news, or one-candle exhaustion.

Scanner output fields

Field	Purpose
Ticker / float / market cap	Liquidity and halt-risk context.
Catalyst type and timestamp	Distinguishes fresh repricing from stale attention.
Gap % from prior close	Measures overnight repricing and potential attention.
Premarket volume and RVOL	Pre-open interest and executable liquidity.
RVOL5 rank at 9:35	Primary stock-in-play confirmation.
ATR14 and ORW/ATR14	Risk/reward and stop distance calibration.
Spread and order-book depth	Execution viability.
Borrow status	Short-side feasibility.
Sector and market alignment	Continuation probability and correlation risk.

5. Stock-in-Play Scoring Model

A scoring model prevents a trader from treating a mediocre breakout like an A+ trade. Scores are not a substitute for execution; they are a throttle for risk.

Component	Max score	A+ guide	Deduct heavily for
Catalyst quality	25	Fresh earnings/guidance, confirmed material event, major sector shock	Vague article, social media, old news, paid promotion
RVOL5 rank	20	Top 20 liquid eligible stock or RVOL5 greater than 3x	Only average opening volume
Liquidity / spread	15	Tight spread, deep book, scalable size	Thin book, hidden spread, low dollar volume
Opening range quality	15	Directional body, limited wick, ORW 8%-30% of ATR14	Huge wick, doji, ORW too wide or too tiny
Technical location	10	Breaks clean premarket/daily level with room to next liquidity pool	Breaks directly into major supply/ demand
Market/sector confirmation	10	Sector confirms or stock is displaying exceptional relative strength	Sector reversing against trade
Execution readiness	5	Orders pre-planned, borrow confirmed, max risk known	No stop calculated or broker rejects short

85-100

A+ setup

Full planned risk unit. Trade only if trigger and acceptance rules align.

70-84

A-/B+ setup

Half risk or wait for retest. Do not chase first extension.

BELOW 70

No trade

Document it. Watch for learning, but protect the account.

Catalyst hierarchy

Tier	Examples	Trading interpretation
Tier 1	Earnings surprise with guidance, regulatory approval/denial, confirmed M&A, major product or contract news, macro-sensitive mega-cap shock	Can drive multi-hour repricing. Eligible for first-break continuation.
Tier 2	Analyst upgrade/downgrade, sector sympathy, litigation update, management change	Tradable if RVOL5 and price action confirm. Prefer tighter risk.
Tier 3	Technical gap only, chatter, low-credibility PR, stale news	Usually no first-break trade. Requires exceptional liquidity and confirmation.

> **Expert rule:** When catalyst quality and relative volume disagree, do not force the trade. A high-quality catalyst with weak volume is not ready. High volume with a weak catalyst is often a trap.

6. Opening Range Construction

The opening range is the first decision boundary. Its job is not to forecast. Its job is to create a clean invalidation zone and reveal whether the first wave of liquidity is directional or indecisive.

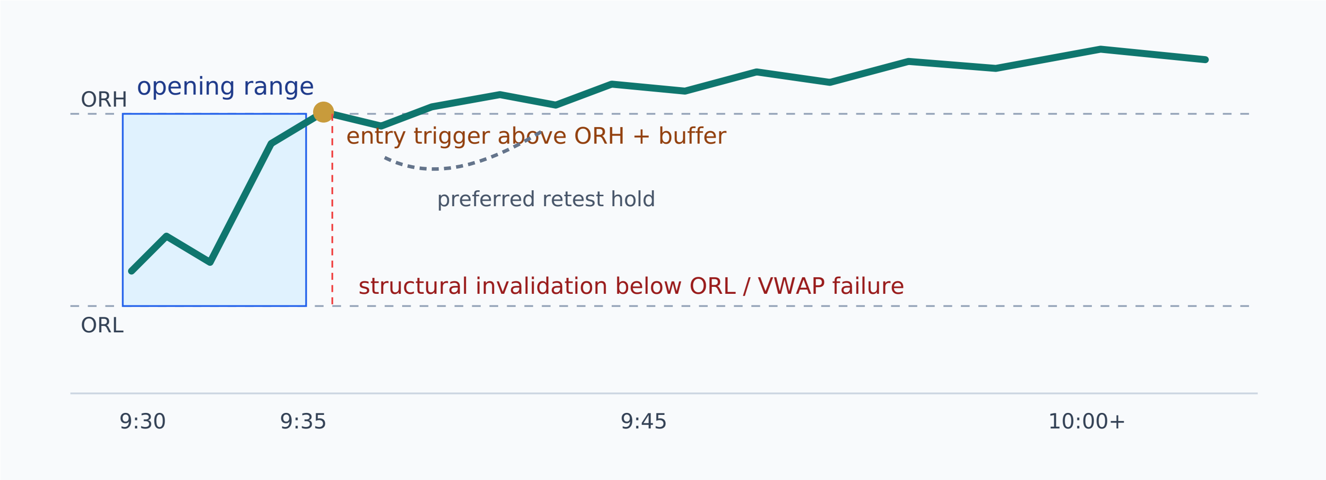
ORH	Opening Range High: highest trade from 9:30:00 through 9:34:59 for the default 5-minute range.
ORL	Opening Range Low: lowest trade from 9:30:00 through 9:34:59.
ORW	Opening Range Width = ORH - ORL.
ORW/ATR	Opening range width divided by 14-day ATR. Used to reject ranges that are too wide or too small.
Acceptance	Price remains outside ORH/ORL after the trigger, with VWAP and tape confirming continuation.

Range quality rules

Condition	Long interpretation	Short interpretation
Opening candle closes near high	Constructive. Buyers controlled the first range.	Only short if later failure is decisive.
Opening candle closes near low	Only long if later failure of sellers is decisive.	Constructive. Sellers controlled the first range.
Large upper/lower wick	Potential rejection. Need retest acceptance.	Potential rejection. Need retest acceptance.
ORW less than 5% ATR14	Risk of noise trigger; require volume and level confirmation.	Risk of noise trigger; require volume and level confirmation.
ORW greater than 35%-40% ATR14	Risk/reward deteriorates; prefer pullback or skip.	Risk/reward deteriorates; prefer pullback or skip.

> **Range-width discipline:** A small range gives cheap risk but more false triggers. A huge range gives clearer information but worse reward/risk. The best ORBs usually balance range clarity with a stop that can be sized rationally.

Visual model



7. Entry Models and Trigger Rules

The playbook uses one primary model and two expert variants. All entries require a completed range, stock-in-play confirmation, and risk calculated before the order is sent.

Model A: A+ Long Opening Range Breakout

Required conditions

- [ ] Tier 1 or strong Tier 2 catalyst.
- [ ] RVOL5 top-ranked or greater than 3x baseline.
- [ ] Opening candle green or constructive after early flush recovery.
- [ ] Price above or reclaiming VWAP before trigger.
- [ ] Break of ORH plus buffer occurs before 10:15.
- [ ] There is room to next premarket/daily resistance for at least 1.5R-2R.

Trigger

```
Long Entry = ORH + buffer
Buffer = max(1-2 ticks, 0.02% of price, 0.10
x 1-min ATR)
Initial stop = structural stop or ATR stop,
defined before entry
```

Use a stop-limit or marketable limit order. If fill quality is poor and price is already extended beyond planned slippage, cancel and wait for a retest.

Model B: A+ Short Opening Range Breakdown

Required conditions

- [ ] Negative catalyst or meaningful gap-down with fresh news.
- [ ] Borrow available or located before trigger.
- [ ] RVOL5 confirms active participation.
- [ ] Opening candle red or failed reclaim of VWAP/OR midpoint.
- [ ] Break of ORL minus buffer occurs before 10:15.
- [ ] There is room to prior support or liquidity zone for at least 1.5R-2R.

Trigger

```
Short Entry = ORL - buffer
Buffer = max(1-2 ticks, 0.02% of price, 0.10
x 1-min ATR)
Initial stop = structural stop or ATR stop,
defined before entry
```

Shorts require additional checks: borrow, locate cost, SSR status, halt risk, and whether a high short-interest squeeze is plausible.

Expert variants

Variant	When to use	Entry tactic	Key risk
Retest continuation	First break occurs too fast or with excess slippage.	Wait for pullback to ORH/ORL or VWAP; enter only if level holds and volume contracts on pullback.	Missing clean trend days that never retest.
Failed-break reversal	Breakout flushes liquidity, immediately rejects, and closes back inside range.	This is a separate reversal strategy. Trade only if you have tested it and size smaller.	Confusing a shakeout with real failure.
Multi-range ORB	Megacaps and ETFs where 5-min range is too noisy.	Use 15-min or 30-min OR; require same stock-in-play logic and market confirmation.	Later entry worsens R/R and may enter after move is mature.

8. Order Tactics and Position Sizing

The ORB trade is execution-sensitive. Small differences in slippage and stop discipline can turn a positive backtest into a negative live strategy.

Order tactics

- **Default:** stop-limit entry with a defined limit offset.
- **Fast tape:** marketable limit, never unbounded market orders in thin stocks.
- **Retest:** passive limit near ORH/ORL if spread is tight and queue risk is acceptable.
- **Bracket:** attach stop immediately or pre-stage a hard stop after fill.
- **Cancel rule:** cancel if price exceeds planned entry by more than permitted slippage.

Position sizing formula

```
Risk per trade ($) = Account equity x risk fraction
Effective risk/share = |Entry - Stop| + expected slippage/share
Shares = floor(Risk per trade / Effective risk/share)
```

Typical risk fraction for an expert intraday strategy is 0.25%-1.00% of account equity per A+ trade, subject to portfolio caps and drawdown state.

Concentration controls

Control	Default	Purpose
Max open ORB positions	3-5	Prevents correlation blowups during opening volatility.
Max sector exposure	2 positions or 1.5R total risk in same sector	Avoids one sector reversal taking all trades.
Max single-name notional	Equal-weight cap or lower if liquidity is weak	Controls gap-through and liquidity risk.
Max entry slippage	10%-20% of planned R, less for scalps	Prevents trading a backtest price that no longer exists.
Max daily loss	2R-3R, hard stop	Protects decision quality during adverse opening conditions.

Never move the stop because position size is too large. Reduce size until the planned stop is financially acceptable.

Worked sizing example

Account equity: $250,000. **Risk fraction:** 0.40%. **Risk budget:** $1,000.

Long entry: $52.20. **Stop:** $51.45. **Expected slippage:** $0.05/share. **Effective risk/share:** $0.80.

```
Shares = floor(1,000 / 0.80) = 1,250 shares
Notional = 1,250 x 52.20 = $65,250
1R = $1,000 before commissions/fees
```

If fill slips to $52.35 but stop remains $51.45, effective risk becomes $0.95 plus slippage. Recalculate or reduce shares.

Common sizing mistakes

Mistake	Consequence	Professional fix
Using fixed share size	Risk fluctuates wildly across tickers.	Size from stop distance and expected slippage.
Ignoring partial fills	Average entry and stop risk are misread.	Recalculate risk after the final fill, then reduce if needed.
Letting wide ranges dictate huge risk	A single false break damages the day.	Skip wide ORW/ATR setups or use retest entries.
Forgetting correlation	Several "independent" trades become one market bet.	Cap sector and index-beta exposure during the opening window.

9. Stop, Exit, and Trade Management

ORB profits come from asymmetric winners. The objective is to cut false breaks quickly and keep a runner only when the stock continues to behave like a true stock in play.

Initial stop framework

Stop type	Use case	Pros	Cons
Structural stop	Below ORL for long; above ORH for short.	Clean thesis invalidation.	Can be too wide after volatile opening bar.
ATR-based stop	Entry minus/plus a tested fraction of ATR14.	Normalizes volatility across stocks.	Can ignore nearby structure.
VWAP failure stop	Breakout loses VWAP and accepts inside range.	Responsive to failed momentum.	Requires fast execution and may whipsaw.
Time stop	No progress after 5-15 minutes.	Frees capital and attention.	May exit before delayed continuation.

Exit protocol

Winner management

- At +1R: take partial or tighten risk according to tested plan.
- At +2R: hold only if price remains outside opening range and above/below VWAP.
- After 10:30: trail using 5-min higher lows/lower highs or VWAP bands.
- Into close: flatten by 15:55 unless the strategy explicitly permits closing-auction exit.

Loser management

- Immediate stop if price re-enters the range and accepts there.
- Immediate stop if catalyst thesis is contradicted by news.
- Reduce if market/sector reverses sharply against trade.
- No averaging down into a failed ORB.

Acceptance rule: For a long, two consecutive 1-minute closes back below ORH after the trigger is a warning. A loss of VWAP plus re-entry into the range is a hard exit signal unless your backtest proves otherwise.

R-multiple plan

Trade state	Action	Reason
`-1R`	Hard exit.	The thesis failed or risk was hit.
`0R` after 10-15 min	Reduce or exit unless tape is building.	Good ORBs usually show progress quickly.
`+1R`	Take 25%-50% or move stop to technical breakeven if tested.	Locks some reward without killing runners.
`+2R`	Trail behind 5-min structure; do not micromanage every tick.	Asymmetric payoffs require runners.
`+3R+`	Hold only if trend conditions remain strong.	Rare winners should fund false breaks.

Management scripts

State	Script
Breakout is clean but extended	"Do not chase beyond planned slippage. Wait for retest or let it go."
Price is hovering at entry	"If no progress by the time stop, reduce or exit. Capital and attention are positions."
Winner reaches +2R early	"Do not turn a trend-day candidate into a scalp unless the tape fails."
Stopped on a valid trade	"Normal loss. No immediate revenge trade in the same name."

10. Advanced Confirmation Overlays

The best traders use overlays to avoid low-quality breakouts, not to rationalize late entries. Each overlay must be defined before the open.

Relative strength overlay

Compare the stock against SPY/QQQ and its sector ETF from 9:30 onward. A long candidate should outperform on green tape or resist pullbacks on red tape. A short candidate should underperform and fail bounces.

```
RS_5m = Stock return since open - Sector ETF return since open
```

VWAP overlay

VWAP is the line between acceptance and rejection. Longs above VWAP with pullbacks holding VWAP are stronger than breakouts extended far above VWAP with no retest.

Premarket level overlay

Best long ORBs clear premarket high or reclaim a key premarket breakdown level. Best short ORBs break premarket low or fail premarket high decisively.

Volume-sequencing overlay

Look for expansion on breakout and contraction on pullback. Expansion on pullback against your trade means the breakout may be failing.

Options-flow overlay

For optionable names, early aggressive call/put flow can support the stock-direction thesis. Treat it as confirmation, not a trigger by itself.

Market-regime overlay

ORB momentum is cleaner when index futures and sector ETFs are trending or stable. Choppy macro days require fewer trades and faster exits.

Overlay hierarchy

Overlay	Use for entry?	Use for sizing?	Use for exit?
Catalyst quality	Yes	Yes	Yes, if contradicted
RVOL5 rank	Yes	Yes	No, except if volume disappears
VWAP acceptance	Yes	Yes	Yes
Sector relative strength	Yes	Yes	Yes
Options flow	Optional	Optional	Only if flow violently reverses
Level 2/order book	Optional	Optional	Useful for slippage and absorption

11. Do-Not-Trade Filters and Failure States

The fastest way to improve ORB expectancy is to stop trading the wrong stocks and the wrong openings. A professional playbook is defined as much by exclusions as by entries.

Hard exclusions

Market and instrument exclusions

- Spread too wide for planned stop and target.
- No confirmed borrow for short setups.
- Recent halt, repeated halt risk, or imminent binary announcement.
- Low dollar volume relative to planned size.
- Opening range is too wide to produce at least 1.5R-2R to next major level.

Behavioral exclusions

- Entering before the range completes because of FOMO.
- Chasing after missing the trigger and widening the stop after entry.
- Taking a second ORB in the same ticker immediately after being stopped.
- Trading a B setup at A+ size.
- Continuing after daily loss limit.

Failure-state diagnostics

Failure state	What it looks like	Action
Liquidity sweep	Price tags ORH/ORL, fills breakout orders, then instantly reverses inside range.	Exit. Do not average. Consider failed-break strategy only if separately tested.
Exhaustion candle	Opening range consumes a large fraction of ATR with huge wick and declining follow-through.	Skip first break. Wait for consolidation or retest.
VWAP rejection	Breakout cannot hold the correct side of VWAP.	Exit or reduce; thesis is not being accepted.
Sector divergence	Stock breaks higher while sector/index rolls over hard.	Require exceptional stock-specific catalyst or stand down.
News ambiguity	Catalyst details are unclear, denied, or contradictory.	Reduce risk or avoid. Ambiguity increases gap-through-stop risk.

Expert risk rule: If the trade requires a perfect fill, perfect stop, and perfect liquidity to make sense, it is not an A+ trade.

12. Systematic Implementation Blueprint

The ORB can be traded discretionarily, but it should be specified like a systematic strategy. Ambiguity belongs in research, not in live execution.

Reference algorithm

```
For each trading day:
  Build eligible universe before open:
    price > price_min
    avg_dollar_volume > adv_min
    ATR14 > atr_min
    spread <= spread_max
    borrow available if shorting

  From 9:30:00 to 9:34:59:
    compute ORH, ORL, ORW, first5_volume, opening_candle_direction

  At 9:35:00:
    RVOL5 = first5_volume / avg_first5_volume_14d
    rank candidates by RVOL5
    keep top N candidates with catalyst_score >= threshold

  For each candidate until entry_cutoff:
    if long conditions true and price >= ORH + buffer:
        enter long with bracket stop
    if short conditions true and price <= ORL - buffer:
        enter short with bracket stop

  Manage:
    stop at invalidation or -1R
    take partial/trail by rule
    flatten by end-of-day
    record all decisions and non-trades
```

Discretionary overlay policy

Discretion is allowed only if pre-defined. Acceptable discretionary reasons include: catalyst contradiction, sudden market-wide macro reversal, borrow failure, unexpected halt risk, or data anomaly. Unacceptable reasons include: fear after a valid trigger, desire to make back losses, widening stop after fill, or changing target because P&L is visible.

Parameter set for first research pass

Parameter	Initial range to test	Notes
Opening range length	5, 10, 15, 30 minutes	Default 5 minutes for high-RVOL stocks; slower assets may need longer ranges.
RVOL lookback	10, 14, 20 sessions	QuantConnect recreation uses prior 14 sessions for first-5-minute volume comparison.[2]
Top candidates	Top 10, 20, 30	Broader lists diversify but lower average quality.
Entry cutoff	9:45, 10:00, 10:15, 10:30	Later entries need stronger confirmation.
Stop model	OR stop, ATR stop, hybrid	Must include slippage and gap-through assumptions.
Exit model	1R partial, trailing stop, close-only, time stop	Exit policy can dominate returns.

Pre-live acceptance test

Gate	Minimum requirement
Historical backtest	Positive out-of-sample expectancy after realistic costs and rejected trades.
Paper trading	At least 20-30 valid signals with execution notes and no unresolved order errors.
Risk rehearsal	Trader can calculate size, place bracket, and cancel orders in under 30 seconds.
Kill switch	Daily R cap and platform/broker controls are tested before live size.

13. Backtesting and Validation Protocol

A professional ORB backtest must be intraday, survivorship-bias-aware, slippage-aware, and timestamp-accurate. Daily bars are insufficient.

Minimum data requirements

Data element	Why it matters
Survivorship-bias-free security master	Delisted and failed stocks affect realistic results.
Minute or tick bars with timestamps	Needed to form opening range and enforce entry cutoff.
Corporate actions adjusted correctly	Splits and dividends must not distort price levels.
Bid/ask or spread model	ORB entries occur during wide-spread periods.
Volume by time-of-day	Needed for RVOL5 and stock-in-play ranking.
Short-borrow assumptions	Short signals cannot assume unlimited borrow.
News/catalyst tags if using catalyst score	A true stock-in-play model needs a reason for abnormal volume.

Validation checklist

Statistical validation

- ☐ Train/test split by time, not random rows.
- ☐ Out-of-sample period includes bull, bear, and high-volatility regimes.
- ☐ Report profit factor, Sharpe, max drawdown, hit rate, average R, tail loss, and exposure.
- ☐ Test parameter sensitivity around chosen values.
- ☐ Segment results by catalyst type, market cap, sector, price, and RVOL decile.

Execution validation

- ☐ Use realistic fill assumptions: no free fills at the high/low of a bar.
- ☐ Include spread, slippage, commissions, fees, and locate costs.
- ☐ Cap participation rate relative to bar volume.
- ☐ Reject trades where stop distance is smaller than plausible slippage.
- ☐ Simulate rejected shorts and halts where possible.

Research caveat: The 2024 ORB results are compelling, but the sample period, symbol universe, cost assumptions, and implementation details matter. Treat them as a research base, not as a plug-and-play guarantee.

Diagnostics to review weekly

Metric	What it reveals
Expectancy by setup grade	Whether the scoring model is actually separating quality.
Slippage by ticker and time	Whether liquidity assumptions are too optimistic.
False-break rate by ORW/ATR	Which range widths should be excluded.
R distribution	Whether winners are large enough to pay for false breaks.
Rule-break P&L	Cost of discretionary violations.

Go-live sign-off

- ☐ Backtest, paper-trade, and first-live-size results are stored in the same journal.
- ☐ Rules for catalyst score, RVOL5, ORW/ATR, entry cutoff, stops, and exits are frozen for the next review cycle.
- ☐ The trader has a written plan for platform outage, data outage, rejected short order, and halted security.
- ☐ All live trades below A- grade are prohibited until the rule-compliant equity curve is positive.

14. Data, Execution, and Infrastructure

An advanced ORB trader needs reliable tools during a compressed time window. The strategy is easy to describe and hard to execute under speed, volatility, and emotion.

Required data stack

- Real-time consolidated quotes and prints.
- Reliable premarket and regular-hours volume.
- VWAP and ATR calculated consistently.
- News feed with timestamps and source quality.
- Sector ETF and index futures context.
- Borrow/locate information for shorts.

Required execution stack

- Hotkeys or order tickets with predefined share-size logic.
- Stop-limit and bracket-order support.
- Ability to cancel quickly if limit offset is breached.
- Real-time P&L hidden or de-emphasized if it affects decisions.
- Automated daily loss lockout if available.
- Execution journal export.

Screen layout for discretionary traders

Panel	Content	Purpose
Watchlist	Score, gap %, RVOL5, spread, ATR, catalyst	Rank only the stocks worth attention.
Main chart	1-min and 5-min candles, VWAP, ORH/ORL, premarket levels	Trigger and management.
Context chart	Daily chart and 30-min chart	Room to run and major supply/demand.
Market/sector	SPY/QQQ/IWM, sector ETF, futures	Macro and correlation filter.
News	Filtered headlines, filings, earnings details	Catalyst validation.
Order monitor	Positions, stops, pending orders, rejected orders	Risk control.

> **Operational principle:** Any task that must happen between 9:35 and 9:37 should be pre-computed or automated. Live calculation under stress creates avoidable errors.

15. Risk Governance and Kill Switches

The difference between professional and amateur ORB trading is not the entry rule. It is loss containment, selective sizing, and process control after the first failed trade.

Risk limits

Limit	Suggested default	Enforcement
Per-trade risk	0.25%-1.00% of equity for A+ setups; lower during drawdown	Calculated before entry.
Daily loss	2R-3R	Hard lockout. No exceptions.
Weekly loss	5R-7R	Reduce size or pause after breach.
Rule-break limit	One rule break triggers size reduction; two triggers shutdown	Journal and review same day.
Consecutive losers	Three losing ORB trades = stop trading ORB that day	Prevents revenge trading in choppy opens.

Drawdown-state sizing

NORMAL

Equity at or near highs; execution metrics stable. Use full planned risk on A+ setups.

CAUTION

Down 3R-5R from peak or slippage deteriorating. Half risk; A+ only; no variants.

DEFENSE

Down 7R+ or rule discipline broken. Pause live trading; review backtest and journal.

Governance checklist

- [ ] Every trade has a pre-entry screenshot or data snapshot with ORH, ORL, RVOL5, catalyst, stop, size, and target.
- [ ] Every exit is tagged: stop, time stop, partial, trail, discretionary, or error.
- [ ] Rule-break P&L is reported separately from strategy P&L.
- [ ] Weekly review separates selection errors, execution errors, management errors, and unavoidable losses.
- [ ] Strategy parameters are changed only after review, not during a losing morning.

> Barber, Lee, Liu, and Odean document large dispersion in day-trader skill, with top-ranked day traders continuing to earn positive returns while bottom-ranked traders continue to lose in later periods.[5] The practical lesson is that process quality matters, and self-assessment must be data-driven.

16. Templates and Worksheets

Use these templates to make the strategy repeatable. The goal is to remove discretion from the parts of the trade that should not be discretionary.

Morning prep worksheet

Field	Entry
Market regime	SPY/QQQ/IWM trend, futures, macro events, volatility state
Top catalysts	Earnings, guidance, news, filings, sector shocks
Do-not-trade list	Wide spread, halt risk, low liquidity, no borrow, ambiguous news
Primary watchlist	Ticker, catalyst tier, gap %, premarket volume, ATR, key levels
Max risk today	Daily R cap, max positions, reduced-size triggers

Trade ticket

Item	Record before entry
Ticker / direction	
Catalyst tier and thesis	
RVOL5 and rank	
ORH / ORL / ORW / ORW-ATR	
Entry / stop / initial target	
Risk per share / shares / total R	
VWAP and sector alignment	
Invalidation statement	"I am wrong if..."

Post-trade review

Question	Tag
Was it a valid stock in play under the written rules?	Selection
Was the entry within planned trigger and slippage?	Execution
Was the initial stop correct and honored?	Risk
Did management follow the R plan?	Management
Was the loss normal or caused by a rule break?	Process
What should change: rule, scanner, size, or nothing?	Review

Best practice: Maintain two equity curves: all trades and rule-compliant trades. If the rule-compliant curve is materially better, the strategy is not the problem - execution discipline is.

17. Bibliography and Source Notes

The playbook blends academic evidence, reproducible research posts, and practitioner process design. URLs are included for source verification and further reading.

1. **Zarattini, Carlo; Barbon, Andrea; Aziz, Andrew.** "A Profitable Day Trading Strategy For The U.S. Equity Market." SSRN, Swiss Finance Institute Research Paper No. 24-98, February 16, 2024. Source notes: paper abstract reports analysis of more than 7,000 U.S. stocks from 2016-2023, focus on 5-minute ORB, Stocks in Play, top-20 portfolio results, Sharpe, and alpha. URL: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4729284
2. **QuantConnect Research.** "Opening Range Breakout for Stocks in Play." December 13, 2024. Source notes: implementation recreation of Zarattini, Barbon, and Aziz; describes liquid universe, price and ATR thresholds, first-5-minute relative-volume definition, top candidates, stop-loss logic, sizing, and 2016 test results. URL: https://www.quantconnect.com/research/18444/opening-range-breakout-for-stocks-in-play/
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4. **Barber, Brad M.; Odean, Terrance.** "All That Glitters: The Effect of Attention and News on the Buying Behavior of Individual and Institutional Investors." The Review of Financial Studies, 2008. Source notes: attention-grabbing stocks include stocks in the news, abnormal volume stocks, and extreme-return stocks. DOI: 10.1093/rfs/hhm079. URL: https://academic.oup.com/rfs/article/21/2/785/1607197
5. **Barber, Brad M.; Lee, Yi-Tsung; Liu, Yu-Jane; Odean, Terrance.** "The Cross-Section of Speculator Skill: Evidence from Day Trading." Journal of Financial Markets, 2014. Source notes: documents large performance dispersion among speculative day traders and persistence among top and bottom groups. URL: https://ideas.repec.org/a/eee/finmar/v18y2014icp1-24.html
6. **Zarattini, Carlo; Aziz, Andrew.** "Can Day Trading Really Be Profitable?" SSRN, April 10, 2023; revised September 22, 2025. Source notes: ORB on QQQ/TQQQ from 2016-2023, annualized alpha and leveraged ETF implementation discussion. URL: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4416622
7. **Barber, Brad M.; Odean, Terrance.** "Trading Is Hazardous to Your Wealth: The Common Stock Investment Performance of Individual Investors." The Journal of Finance, 2000. Source notes: active individual investors underperform passive market return in a large brokerage sample. URL: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=219228
8. **SMB Training.** "15 Best Day Trading Stocks YTD (2023)." August 9, 2023. Source notes: practitioner/proprietary-desk style description of Stocks in Play criteria such as catalyst, liquidity, range, and RVOL. URL: https://www.smbtraining.com/blog/15-best-day-trading-stocks-ytd-2023

> **Final implementation principle:** The ORB strategy is only as good as its selection discipline. Trade the opening range only when the stock deserves to be traded, the range is clean, and the risk is predetermined.